

BalanceFlow Backend API Documentation

Generated from the local Spring Boot source. Base URL: <http://localhost:8080>

Backend	Spring Boot 3.5.14, Java 17, JPA, PostgreSQL, Redis, Mail, JWT
Auth	Stateless JWT. Use Authorization: Bearer <token> on protected APIs.
Public	/api/auth/**, /test-token, /redis-test, /mail-test, /error
Protected	Everything else requires a valid JWT.
Error style	Most invalid cases throw RuntimeException or IllegalArgumentException and use Spring's default error response.

Frontend Flow

- Register or login, then save token from /api/auth/login and attach Authorization: Bearer <token> to protected requests.
- Forgot password flow: send OTP, verify OTP, then reset password within the 10 minute verified window.
- After login, load profile, dashboard summary, my groups, and notifications for the home screen.
- Create a group with POST /api/groups. The backend makes the creator OWNER and logs activity.
- Use search users, then add a member by email. Refresh members, activities, and notifications.
- Group detail should load group details, members, expenses, balances, suggested settlements, settlement history, and activities.
- Adding an expense only needs title and amount. Backend splits equally across all current members and notifies non-payers.
- Suggested settlements are calculated from current expense balances. Creating a settlement records history/activity/notification, but current balance calculation still uses expenses only.
- Personal categories and transactions are separate from group expenses. Create categories first, then transactions with categoryId.
- Show owner-only UI for remove member and delete group. Owner cannot leave while other members exist.

Implementation Notes

Topic	Note
JWT	Controllers often manually read Authorization and extract the email.
Authorization gaps	Spring checks authentication globally, but several group/expense endpoints do not verify group membership in service logic.
Admin stats	/api/dashboard/stats is authenticated but has no role check.
Expense split	Only equal split across all group members is supported.
Settlements	Stored settlement records do not currently reduce computed balances.
Categories	Categories are global records, not per-user.
Validation	Most DTOs have no bean validation annotations; service/runtime errors handle invalid input.

Authentication

Register

Method	API link	Auth required
POST	http://localhost:8080/api/auth/register	No

Notes: Creates user. Email must be unique.

Request body

```
{
  "fullName": "Heta Karani",
  "email": "heta@example.com",
  "password": "secret123"
}
```

Success response

```
{
  "id": "uuid",
  "fullName": "Heta Karani",
  "email": "heta@example.com"
}
```

Login

Method	API link	Auth required
POST	http://localhost:8080/api/auth/login	No

Notes: Store token and send Authorization: Bearer <token>. JWT expires after 24 hours.

Request body

```
{
  "email": "heta@example.com",
  "password": "secret123"
}
```

Success response

```
{
  "email": "heta@example.com",
  "token": "jwt-token",
  "fullName": "Heta Karani"
}
```

Send OTP

Method	API link	Auth required
POST	http://localhost:8080/api/auth/send-otp	No

Notes: OTP is stored in Redis for 5 minutes and sent by email.

Request body

```
{
  "email": "heta@example.com"
}
```

Success response

```
{
  "success": true,
  "message": "OTP sent successfully"
}
```

Verify OTP

Method	API link	Auth required
POST	http://localhost:8080/api/auth/verify-otp	No

Notes: Creates verified:<email> in Redis for 10 minutes.

Request body

```
{
  "email": "heta@example.com",
  "otp": "123456"
}
```

Success response

```
{
  "success": true,
  "message": "OTP verified successfully"
}
```

Reset Password

Method	API link	Auth required
POST	http://localhost:8080/api/auth/reset-password	No

Notes: Requires successful OTP verification first.

Request body

```
{
  "email": "heta@example.com",
  "newPassword": "newSecret123"
}
```

Success response

```
{
  "success": true,
  "message": "Password reset successfully"
}
```

Profile and Users

Get Profile

Method	API link	Auth required
GET	http://localhost:8080/api/profile	Yes

Request body

No body

Success response

```
{
  "id": "uuid",
  "fullName": "Heta Karani",
  "email": "heta@example.com",
  "joinedAt": "2026-06-19T12:00:00"
}
```

Update Profile

Method	API link	Auth required
PATCH	http://localhost:8080/api/profile	Yes

Request body

```
{
  "fullName": "Heta K"
}
```

Success response

```
{
  "id": "uuid",
  "fullName": "Heta K",
  "email": "heta@example.com",
  "joinedAt": "2026-06-19T12:00:00"
}
```

Get User Profile

Method	API link	Auth required
GET	http://localhost:8080/api/users/profile	Yes

Notes: Parallel profile endpoint without joinedAt.

Request body

No body

Success response

```
{
  "id": "uuid",
  "fullName": "Heta K",
  "email": "heta@example.com"
}
```

Update User Profile

Method	API link	Auth required
PUT	http://localhost:8080/api/users/profile	Yes

Request body

```
{
  "fullName": "Heta K"
}
```

Success response

```
{
  "id": "uuid",
  "fullName": "Heta K",
  "email": "heta@example.com"
}
```

Change Password

Method	API link	Auth required
POST	http://localhost:8080/api/users/change-password	Yes

Request body

```
{
  "oldPassword": "secret123",
  "newPassword": "newSecret123"
}
```

Success response

```
{
  "message": "Password changed successfully"
}
```

Groups

Create Group

Method	API link	Auth required
POST	http://localhost:8080/api/groups	Yes

Notes: Creator becomes OWNER. Activity is logged.

Request body

```
{
  "name": "Goa Trip",
  "description": "Trip expenses"
}
```

Success response

```
{
  "id": "uuid",
  "name": "Goa Trip",
  "description": "Trip expenses",
  "createdBy": "Heta Karani"
}
```

List My Groups

Method	API link	Auth required
GET	http://localhost:8080/api/groups/my-groups	Yes

Notes: GET /api/groups maps to same handler.

Request body

No body

Success response

```
[
  {
    "id": "uuid",
    "name": "Goa Trip",
    "description": "Trip expenses",
    "role": "OWNER"
  }
]
```

Add Member

Method	API link	Auth required
POST	http://localhost:8080/api/groups/{groupId}/members	Yes

Notes: Adds existing registered user as MEMBER. Creates activity and notification.

Request body

```
{
  "email": "friend@example.com"
}
```

Success response

Member added successfully

Get Group Details

Method	API link	Auth required
GET	http://localhost:8080/api/groups/{groupId}	Yes

Request body

No body

Success response

```
{
  "id": "uuid",
  "name": "Goa Trip",
  "description": "Trip expenses",
  "memberCount": 3,
  "totalExpenses": 5,
  "createdBy": "Heta Karani"
}
```

Get Group Members

Method	API link	Auth required
GET	http://localhost:8080/api/groups/{groupId}/members	Yes

Request body

No body

Success response

```
[
  {
    "id": "uuid",
    "fullName": "Heta Karani",
    "email": "heta@example.com",
    "role": "OWNER"
  }
]
```

Remove Member

Method	API link	Auth required
DELETE	http://localhost:8080/api/groups/{groupId}/members/{userId}	Yes

Notes: Only OWNER can remove members.

Request body

No body

Success response

Member removed successfully

Leave Group

Method	API link	Auth required
POST	http://localhost:8080/api/groups/{groupId}/leave	Yes

Notes: OWNER cannot leave while other members exist.

Request body

No body

Success response

Left group successfully

Delete Group

Method	API link	Auth required
DELETE	http://localhost:8080/api/groups/{groupId}	Yes

Notes: Only OWNER can delete. Cascades group data.

Request body

No body

Success response

Group deleted successfully

Expenses, Balances, Settlements, Activities

Create Expense

Method	API link	Auth required
POST	http://localhost:8080/api/groups/{groupId}/expenses	Yes

Notes: Paid by current user. Equal split across all current group members.

Request body

```
{
  "title": "Dinner",
  "amount": 1200.0
}
```

Success response

```
{
  "id": "uuid",
  "title": "Dinner",
  "amount": 1200.0,
  "paidBy": "Heta Karani",
  "participants": [
    {
      "fullName": "Heta Karani",
      "shareAmount": 400.0
    },
    {
      "fullName": "Friend",
      "shareAmount": 400.0
    }
  ]
}
```

List Group Expenses

Method	API link	Auth required
GET	http://localhost:8080/api/groups/{groupId}/expenses	Yes

Request body

No body

Success response

```
[
  {
    "id": "uuid",
    "title": "Dinner",
    "amount": 1200.0,
    "paidBy": "Heta Karani",
    "participants": [
      {
        "fullName": "Friend",
        "shareAmount": 400.0
      }
    ]
  }
]
```

Update Expense

Method	API link	Auth required
PATCH	http://localhost:8080/api/groups/{groupId}/expenses/{expenseId}	Yes

Notes: If amount changes, participant shares are recalculated equally.

Request body

```
{
  "title": "Dinner updated",
  "amount": 1500.0
}
```

Success response

```
{
  "id": "uuid",
  "title": "Dinner updated",
  "amount": 1500.0,
  "paidBy": "Heta Karani",
  "participants": [
    {
      "fullName": "Friend",
      "shareAmount": 500.0
    }
  ]
}
```

Get Expense Details

Method	API link	Auth required
GET	http://localhost:8080/api/groups/{groupId}/expenses/{expenseId}	Yes

Request body

No body

Success response

```
{
  "id": "uuid",
  "title": "Dinner",
  "amount": 1200.0,
  "paidBy": "Heta Karani",
  "groupId": "uuid",
  "groupName": "Goa Trip",
  "participants": [
    {
      "fullName": "Friend",
      "shareAmount": 400.0
    }
  ]
}
```

Delete Expense

Method	API link	Auth required
DELETE	http://localhost:8080/api/groups/{groupId}/expenses/{expenseId}	Yes

Request body

No body

Success response

Expense deleted successfully

Get Group Balances

Method	API link	Auth required
GET	http://localhost:8080/api/groups/{groupId}/balances	Yes

Notes: Map keys are full names.

Request body

No body

Success response

```
{
  "Heta Karani": {
    "paid": 1200.0,
    "owes": 400.0,
    "net": 800.0
  },
  "Friend": {
    "paid": 0.0,
    "owes": 400.0,
    "net": -400.0
  }
}
```

Get Suggested Settlements

Method	API link	Auth required
GET	http://localhost:8080/api/groups/{groupId}/settlements	Yes

Notes: Calculated suggestions only, not saved history.

Request body

No body

Success response

```
[
  {
    "from": "Friend",
    "to": "Heta Karani",
    "amount": 400.0
  }
]
```

Create Settlement

Method	API link	Auth required
POST	http://localhost:8080/api/groups/{groupId}/settlements	Yes

Notes: Validates amount > 0, payer != receiver, and both users are members. Creates activity and notification.

Request body

```
{
  "payerId": "uuid-of-debtor",
  "receiverId": "uuid-of-creditor",
  "amount": 400.0
}
```

Success response

```
{
  "id": "uuid",
  "payer": "Friend",
  "receiver": "Heta Karani",
  "amount": 400.0,
  "settledAt": "2026-06-19T12:00:00"
}
```

Settlement History

Method	API link	Auth required
GET	http://localhost:8080/api/groups/{groupId}/settlements/history	Yes

Request body

No body

Success response

```
[
  {
    "id": "uuid",
    "payer": "Friend",
    "receiver": "Heta Karani",
    "amount": 400.0,
    "settledAt": "2026-06-19T12:00:00"
  }
]
```

Group Activities

Method	API link	Auth required
GET	http://localhost:8080/api/groups/{groupId}/activities	Yes

Request body

No body

Success response

```
[
  {
    "id": "uuid",
    "action": "Expense Added: Dinner by Heta Karani",
    "createdBy": "Heta Karani",
    "createdAt": "2026-06-19T12:00:00"
  }
]
```

Categories and Transactions

Create Category

Method	API link	Auth required
POST	http://localhost:8080/api/categories	Yes

Request body

```
{
  "name": "Food",
  "type": "EXPENSE"
}
```

Success response

```
{
  "id": "uuid",
  "name": "Food",
  "type": "EXPENSE"
}
```

List Categories

Method	API link	Auth required
GET	http://localhost:8080/api/categories	Yes

Request body

No body

Success response

```
[
  {
    "id": "uuid",
    "name": "Food",
    "type": "EXPENSE"
  }
]
```

Update Category

Method	API link	Auth required
PUT	http://localhost:8080/api/categories/{id}	Yes

Request body

```
{
  "name": "Salary",
  "type": "INCOME"
}
```

Success response

```
{
  "id": "uuid",
  "name": "Salary",
  "type": "INCOME"
}
```

Delete Category

Method	API link	Auth required
DELETE	http://localhost:8080/api/categories/{id}	Yes

Request body

No body

Success response

```
{
  "message": "Category deleted successfully"
}
```

Create Transaction

Method	API link	Auth required
POST	http://localhost:8080/api/transactions	Yes

Request body

```
{
  "amount": 500.0,
  "note": "Lunch",
  "transactionDate": "2026-06-19",
  "categoryId": "uuid"
}
```

Success response

```
{
  "id": "uuid",
  "amount": 500.0,
  "note": "Lunch",
  "transactionDate": "2026-06-19",
  "categoryName": "Food",
  "categoryType": "EXPENSE"
}
```

List Transactions

Method	API link	Auth required
GET	http://localhost:8080/api/transactions	Yes

Notes: Optional filter: ?categoryId=<uuid>.

Request body

No body

Success response

```
[
  {
    "id": "uuid",
    "amount": 500.0,
    "note": "Lunch",
    "transactionDate": "2026-06-19",
    "categoryName": "Food",
    "categoryType": "EXPENSE"
  }
]
```

Update Transaction

Method	API link	Auth required
PUT	http://localhost:8080/api/transactions/{id}	Yes

Notes: Only owner can update.

Request body

```
{
  "amount": 650.0,
  "note": "Dinner",
  "transactionDate": "2026-06-19",
  "categoryId": "uuid"
}
```

Success response

```
{
  "id": "uuid",
  "amount": 650.0,
  "note": "Dinner",
  "transactionDate": "2026-06-19",
  "categoryName": "Food",
  "categoryType": "EXPENSE"
}
```

Delete Transaction

Method	API link	Auth required
DELETE	http://localhost:8080/api/transactions/{id}	Yes

Notes: Only owner can delete.

Request body

No body

Success response

```
{  
  "message": "Transaction deleted successfully"  
}
```

Dashboard, Search, Notifications, Test Helpers

Dashboard Summary

Method	API link	Auth required
GET	http://localhost:8080/api/dashboard	Yes

Request body

No body

Success response

```
{
  "totalGroups": 2,
  "totalExpenses": 8,
  "totalSpent": 3500.0,
  "totalOwed": 900.0
}
```

Admin Stats

Method	API link	Auth required
GET	http://localhost:8080/api/dashboard/stats	Yes

Notes: No role check is implemented.

Request body

No body

Success response

```
{
  "totalGroups": 12,
  "totalExpenses": 45,
  "totalSettlements": 7,
  "totalUsers": 20
}
```

Search Groups

Method	API link	Auth required
GET	http://localhost:8080/api/search/groups?q=goa	Yes

Request body

No body

Success response

```
[
  {
    "id": "uuid",
    "name": "Goa Trip",
    "description": "Trip expenses"
  }
]
```


Search Users

Method	API link	Auth required
GET	http://localhost:8080/api/search/users?q=heta	Yes

Request body

No body

Success response

```
[
  {
    "id": "uuid",
    "fullName": "Heta Karani",
    "email": "heta@example.com"
  }
]
```

List Notifications

Method	API link	Auth required
GET	http://localhost:8080/api/notifications	Yes

Request body

No body

Success response

```
[
  {
    "id": "uuid",
    "title": "New Expense",
    "message": "You were added to expense: Dinner in group Goa Trip",
    "isRead": false,
    "createdAt": "2026-06-19T12:00:00"
  }
]
```

Mark Notification Read

Method	API link	Auth required
PATCH	http://localhost:8080/api/notifications/{id}/read	Yes

Request body

No body

Success response

Empty response body

Mark All Notifications Read

Method	API link	Auth required
PATCH	http://localhost:8080/api/notifications/read-all	Yes

Request body

No body

Success response

Empty response body

JWT Test Token

Method	API link	Auth required
GET	http://localhost:8080/test-token	No

Notes: Development helper.

Request body

No body

Success response

jwt-token-for-het@gmail.com

Redis Test

Method	API link	Auth required
GET	http://localhost:8080/redis-test	No

Notes: Development helper.

Request body

No body

Success response

heta

Mail Test

Method	API link	Auth required
GET	http://localhost:8080/mail-test	No

Notes: Development helper.

Request body

No body

Success response

Email sent

Suggested Frontend Screen Map

Screen	Primary APIs
Login/Register	POST /api/auth/register, POST /api/auth/login
Forgot Password	POST /api/auth/send-otp, /verify-otp, /reset-password
Dashboard	GET /api/profile, /api/dashboard, /api/groups/my-groups, /api/notifications
Groups List	GET /api/groups/my-groups, POST /api/groups
Group Detail	GET group details, members, expenses, balances, settlements, history, activities
Expense Form	POST/PATCH /api/groups/{groupId}/expenses
Settle Up	GET suggested settlements, POST saved settlement
Personal Ledger	Categories plus transactions APIs
Search/Add People	GET /api/search/users?q=..., POST group member
Notifications	GET /api/notifications, PATCH read/read-all